Palm Oil Retention & Melting System

Architectural Revision II — Thermofluid Analysis and Controls Engineering

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Based on the *VCH Sizing Framework, Second Edition*
A Unified Design Methodology for Submerged Hydronic Coil Systems

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This document presents a minor revision to the Architectural Revision II (ADA) design: the coil bore is reduced from 2 inch to 1.5 inch across every tank, the branch valve count drops from 8 to 6, and the need array is expanded so that all three priorities (P1, P2, P3) can be genuinely concurrent.

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Executive Summary

This document presents a minor revision to the Architectural Revision II (ADA) design: the coil bore across every tank is reduced from 2 inch to 1.5 inch, the branch valve count drops from 8 to 6, and — most significantly — the need array is expanded so that all **three** priorities (P1, P2, P3) can be genuinely concurrent, not just two at a time. Two of the three branches now serve the T_{10} melting duty continuously and permanently; the third branch is shared between T_5 (P1) and the PO tank (P2), with the trunk itself capped at a safe $1.2\,\mathrm{m\,s^{-1}}$ — meaning three-way concurrency throttles every active branch below its individual design flow rather than starving any one process outright. This document works the full thermofluid consequence of that change from first principles, quantifies the resulting throughput, checks the system's peak instantaneous power draw against the auxiliary array's rated capacity, restates the ADA architecture as it now stands, and records the current PLC valve mapping.

Headline Results

Metric	Result
Coil area lost to the bore reduction (all tanks)	−20.0%
T_5 (P1) daily duty period	4.47 hr (18.6%)
PO tank (P2) daily duty period	0.71 hr (2.96%)
Shared-branch active fraction of the day	21.0%
Palm oil melted per day (P3)	≈4453 kg/day
Combined system throughput (P3 + PO tank)	≈4953 kg/day
Fulfilment vs. 2 t/day mandate (melt / combined)	≈222.7% / ≈247.7%
Total daily thermal energy demand	≈990.7 MJ
Average continuous power demand	≈11.47 kW
Peak instantaneous demand (worst-case)	**≈36.61 kW**
Auxiliary array rated capacity	36.0 kW

Key Findings

- **The bore reduction, taken alone, costs the low-NTU duties (T_5 and T_{10} melting) close to their full 20 % area loss** — delivered heat tracks lost area almost 1:1 when $NTU \ll 1$. The PO tank, sitting at meaningfully higher NTU, is structurally buffered and loses proportionally less.
- **The three-way flow split, isolated from the bore reduction, costs remarkably little** — every duty retains 98–99 % of its own dual-branch delivery rate when throttled to three-way concurrent flow.
- **Combined throughput falls by** ≈8.6 % relative to the prior (2-inch, two-priority) revision, but still clears the original 2 t/day mandate by ≈**2.5**×.
- **At the single worst-case instant where every process is simultaneously at its coldest, highest-draw condition, peak demand** (≈36.61 kW) **marginally**

exceeds the auxiliary array's rated 36 kW — a genuine, if narrow, finding worth carrying into the electrical design phase (Section 9).

1 Scope and Motivation of the Minor Revision

Architectural Revision II fixed the coil at 2 inch bore across every tank and allowed the need array to serve at most two branches at once. Two further hardware changes are introduced here:

1. **Coil bore reduced to** 1.5 inch across every tank (T_5, T_{10}, PO), coil lengths unchanged (10 m for T_5, 30 m for T_{10}/PO).
2. **Branch valve count reduced from 8 to 6**, and the need array is expanded so that **all three priorities can draw simultaneously**, not just two — the trunk is capped at a safe 1.2 m s^{-1} rather than being sized to guarantee every branch its individual full design flow under three-way concurrency.

The second change is the more consequential one architecturally: it retires the three-way "discard the lowest priority" contention rule from Revision II's PLC arbiter (Section 14 of that document) in favour of genuine three-way service, at a throttled flow rate, whenever all three priorities have live demand at once.

2 Revised Coil Geometry

Table 1. Coil geometry, 2-inch (prior revision) vs. 1.5-inch (this revision).

Parameter	T_5 (10 m)		T_{10}/PO (30 m)	
	2-inch	1.5-inch	2-inch	1.5-inch
d_o (mm)	60.33	48.26	60.33	48.26
d_i (mm)	54.79	42.72	54.79	42.72
d_o/d_i	1.1011	1.1297	1.1011	1.1297
R_{wall} ($\times 10^{-4}$ m^2.K.W^{-1})	1.782	1.805	1.782	1.805
Coil surface area (m^2)	1.895	**1.516**	5.685	**4.549**
Area change	−20.0%		−20.0%	

Assumption. T_5's coil is treated as changing from 2 inch to 1.5 inch alongside T_{10}/PO, consistent with the most recent instruction received. If T_5's coil is in fact unchanged, only the T_{10}/PO columns of every downstream table apply to it.

3 Trunk Reconfiguration: True Three-Way Concurrent Flow

3.1 Flow Split at the Trunk's Own Velocity Ceiling

The 1.5-inch trunk (area $1.433 \times 10^{-3}\,\mathrm{m}^2$) is capped at its own safe rated velocity of $1.2\,\mathrm{m\,s^{-1}}$ — the same numeric limit previously applied to the 1-inch branches individually, now applied to the trunk as a whole. Under three-way concurrency, the trunk runs at its own ceiling and the branches share what it can deliver:

$$F_{\text{trunk}} = 1.433 \times 10^{-3} \times 1.2 = 1.7196 \times 10^{-3}\,\mathrm{m^3\,s^{-1}} \tag{1}$$

$$F_{\text{branch,triple}} = \frac{F_{\text{trunk}}}{3} = 5.732 \times 10^{-4}\,\mathrm{m^3\,s^{-1}} \;\Rightarrow\; V_{\text{branch,triple}} \approx 0.944\,\mathrm{m\,s^{-1}} \tag{2}$$

Single- and dual-branch operation are unaffected by this change. A single active branch or two active branches together still draw at most $2 \times 7.283 \times 10^{-4}\,\mathrm{m^3\,s^{-1}}$, well inside the trunk's own $1.2\,\mathrm{m\,s^{-1}}$ ceiling — those branches continue to receive their full individual $1.2\,\mathrm{m\,s^{-1}}$ design rate exactly as in Revision II. The throttled regime worked through in this document applies **only** to the moment all three branches draw simultaneously.

Note on a minor cross-reference inconsistency. The branch flow area used above ($6.069 \times 10^{-4}\,\mathrm{m}^2$) traces to the project's own hand-derived trunk/branch sizing notes; the original base VCH proposal's 1-inch supply pipe basis instead implies $\approx 5.575 \times 10^{-4}\,\mathrm{m}^2$ (26.64 mm ID). This $\approx 9\%$ area discrepancy affects only the *reported branch velocity* figure — every thermal result in this document is derived from **volumetric flow rate** at the trunk split, which is unaffected by which branch-ID convention is used for the velocity label. Worth reconciling against as-built pipe specification, but it does not change any C_w, NTU, or throughput figure below.

3.2 Capacity Rate Under Both Regimes

$$C_{w,\text{single/dual}} = 2740.7\,\mathrm{W\,K^{-1}} \text{ (unchanged, Revision II basis)} \tag{3}$$

$$\dot{m}_{w,\text{triple}} = 977.8 \times 5.732 \times 10^{-4} = 0.5605\,\mathrm{kg\,s^{-1}} \;\Rightarrow\; C_{w,\text{triple}} = 2348.6\,\mathrm{W\,K^{-1}} \tag{4}$$

A 14.3 % drop in C_w under three-way concurrency, relative to the single/dual regime.

4 Water-Side Hydraulics at the New Bore

Water at 70 °C: $\rho = 977.8\,\mathrm{kg\,m^{-3}}$, $C_p = 4190\,\mathrm{J\,kg^{-1}\,K^{-1}}$, $\mu = 4.04 \times 10^{-4}\,\mathrm{Pa\,s}$, $k = 0.663\,\mathrm{W\,m^{-1}\,K^{-1}}$, $\mathrm{Pr} = 2.55$.

Table 2. Reynolds number and Gnielinski friction factor, 1.5-inch bore.

Parameter	Single/dual ($\dot{m} = 0.654\,\mathrm{kg\,s^{-1}}$)	Triple ($\dot{m} = 0.5605\,\mathrm{kg\,s^{-1}}$)
Re_{coil}	48,251	41,349
$f = (0.79 \ln Re - 1.64)^{-2}$	0.02113	0.02190
Nu_{str}	203.2	178.75

Table 3. Inner heat transfer coefficient by tank and flow regime, 1.5-inch bore.

Tank	Single/dual (W/m².K)	Triple (W/m².K)
$h_{i,T10/PO}$ ($D_{\mathrm{helix}} = 2.072\,\mathrm{m}$)	3384	2978
$h_{i,T5}$ ($D_{\mathrm{helix}} = 1.711\,\mathrm{m}$)	3432	3020

5 Revised Overall U-Values

$$\frac{1}{U} = \frac{1}{h_i} \cdot \frac{d_o}{d_i} + R_{f,i} \cdot \frac{d_o}{d_i} + R_{\mathrm{wall}} + R_{f,o} + \frac{1}{h_o} \tag{5}$$

using the same fouling basis as every prior revision ($R_{f,i} = 1.0 \times 10^{-4}$, $R_{f,o} = 9.0 \times 10^{-4}$ $\mathrm{m^2.K.W^{-1}}$).

Table 4. Overall U-values, 1.5-inch bore, all duties and regimes.

Duty	h_o (W/m².K)	U single/dual (W/m².K)	U triple (W/m².K)
T_5 (liquid)	138.9	115.59	115.13
PO tank (liquid)	138.9	**115.55**	115.08
T_{10} melt (solid)	35	33.28	33.23

Correction to an earlier working figure. An intermediate chat exchange during this revision's development stated the PO tank's single/dual-branch U-value as $128.7\,\mathrm{W\,m^{-2}\,K^{-1}}$ — an arithmetic slip. The correct recomputed value, shown above, is $115.55\,\mathrm{W\,m^{-2}\,K^{-1}}$, essentially identical to T_5's own value (both liquid duties, same bore, same h_o, similar h_i). This correction affects **only** the single/dual-branch comparison row for the PO tank; every duty-period, throughput, and energy figure elsewhere in this document was already built from the correct triple-branch PO figure and is unaffected.

6 NTU, Effectiveness, and Delivered Heat Rate

$$NTU = \frac{UA}{C_w}, \qquad \varepsilon = 1 - e^{-NTU}, \qquad \varepsilon C_w = \text{delivered rate per unit } \Delta T \tag{6}$$

Table 5. Full comparison: 2-inch (Revision II baseline) vs. 1.5-inch, both flow regimes. All εC_w values in W/K.

Duty	εC_w, 2" baseline	εC_w, 1.5" single/dual	Δ vs. baseline	εC_w, 1.5" triple	Δ vs. 1.5" single/dual
T_5	205.3	169.75	−17.3%	168.28	−0.9%
PO tank	570.5	478.65	−16.1%	469.6	−1.9%
T_{10} (solid)	181.5	147.38	−18.8%	146.42	−0.7%

Two independent effects, cleanly separable. Reading left-to-right within the 1.5-inch columns isolates the **bore reduction** (baseline → single/dual, roughly 16–19 % down across all three duties — tracking the 20 % area loss closely for the low-NTU duties). Reading single/dual → triple isolates the **three-way flow-split penalty** alone, and it is small — under 2 % for every duty. The two effects should not be conflated: the coil now delivers meaningfully less than before regardless of branch count, but adding the third concurrent branch on top of that already-reduced coil costs almost nothing further.

7 The Need-Array Expansion to True Three-Way Concurrency

7.1 Architecture

Two of the three branches are now permanently and continuously dedicated to T_{10} palm oil melting (P3) — there is no longer a scenario in which P3 is fully idle. The third branch is shared between P1 (T_5) and P2 (PO tank), exactly as arbitrated in Revision II's need array. The structural consequence: **any time P1 or P2 has live demand, the trunk is necessarily supplying all three branches at once** — P1 and P2 never see the better single/dual-branch rate; they always operate under the throttled triple-branch condition. Only the two melting branches ever experience the untouched, full single/dual-branch rate, and only during the majority of the day when neither P1 nor P2 has any demand.

This retires the "discard the lowest priority" three-way contention rule from Revision II's PLC arbiter (Section 14, Revision II document). That rule existed specifically to resolve genuine three-way contention when only two branches were physically available. It is now obsolete: three-way contention is the trunk's normal operating condition whenever P1 or P2 is active, and it is resolved by *flow-sharing*, not by *discarding* a priority.

7.2 Timing

Using T_5's daily demand basis (4 t/day, unchanged) and the PO tank's (500 kg/day, unchanged), evaluated **entirely at the triple-branch rate** since that is the only regime P1/P2 ever see:

$$t_{P1} = \frac{74.4 \times 10^6}{168.28 \times 27.5} \approx 16\,077\,\text{s} \approx \mathbf{4.47\ hr}\ (\mathbf{18.6}\%) \tag{7}$$

$$t_{P2} = \frac{51.7 \times 10^6}{469.6 \times 43} \approx 2561\,\text{s} \approx \mathbf{0.71\ hr}\ (\mathbf{2.96}\%) \tag{8}$$

$$t_{\text{shared,active}} \approx t_{P1} + t_{P2} - \frac{t_{P1} \times t_{P2}}{24} \approx 4.47 + 0.71 - 0.13 \approx \mathbf{5.05\ hr}\ (\mathbf{21.0}\%) \tag{9}$$

$t_{\text{shared,idle}} = 24 - 5.05 = \mathbf{18.95\ hr}\ (\mathbf{79.0}\%)$ — both melting branches at full single/dual rate (10)

8 P3 Output Under Mixed-Regime Operation

$$Q_{\text{full}}(\text{per tank}, 36\,^\circ\text{C}) = 147.38 \times 34 = 5011\,\text{W} \tag{11}$$

$$Q_{\text{reduced}}(\text{per tank}, 36\,^\circ\text{C}) = 146.42 \times 34 = 4978\,\text{W} \tag{12}$$

$$P_{P3} = 0.790 \times (2 \times 5011) + 0.210 \times (2 \times 4978) \approx 10.007\,\text{W} \tag{13}$$

$$E_{P3} \approx 864.6\,\text{MJ/day} \;\Rightarrow\; \boxed{m_{P3} = \frac{864.6 \times 10^6}{194{,}200} \approx \mathbf{4453\ kg/day}} \tag{14}$$

9 Maximum Instantaneous Wattage Analysis

9.1 Motivation

Every duty-period and throughput figure above is a *time-averaged* result. Sizing the auxiliary electric array, by contrast, requires knowing the **worst-case simultaneous** demand — the single instant every process could plausibly draw its individual maximum at once.

9.2 Worst-Case Coincidence

The most conservative physically-plausible instant is: T_5 has just triggered at 40 °C (its coldest, highest-draw condition); the PO tank has just triggered at 24 °C (likewise); and both melting branches are simultaneously at the 36 °C melt point (their own highest-draw condition). Since this instant necessarily involves both P1 and P2 active, both melting branches are also in the throttled triple-branch regime at this moment.

$$Q_{T5,\max} = 168.28 \times (70 - 40) = 5048\,\text{W} \tag{15}$$

$$Q_{PO,\max} = 469.6 \times (70 - 24) = 21.602\,\text{W} \tag{16}$$

$$Q_{T10,\max} \times 2 = (146.42 \times 34) \times 2 = 9957\,\text{W} \tag{17}$$

$$\boxed{P_{\text{peak}} = 5048 + 21{,}602 + 9957 \approx \mathbf{36{,}607\ W}\ (\mathbf{36.61\ kW})} \tag{18}$$

This exceeds the auxiliary array's rated 36 kW **capacity by** ≈1.7 %**.** This is a compounded worst-case figure — every process simultaneously at its own individual coldest condition — and is deliberately conservative. In practice, the 300 L active thermal buffer mass established in the base VCH proposal would absorb a brief coincidence of this kind as a small, temporary temperature dip rather than an outright shortfall. However, the margin is narrow enough (1.7 % over) that it is worth carrying into the electrical design phase as an explicit check, rather than assuming the existing 36 kW array has headroom against every possible operating condition.

10 Full Before/After Comparison

Table 6. Architectural Revision II vs. this minor revision.

Metric	Revision II (2”, 2-priority)	This revision (1.5”, 3-priority)
T_5 duty period	3.67 hr (15.3%)	4.47 hr (18.6%)
PO tank duty period	0.585 hr (2.4%)	0.71 hr (2.96%)
P3 melt output	4919.8 kg/day	4453 kg/day (−9.5%)
Combined throughput	5419.8 kg/day	4953 kg/day (−8.6%)
Fulfilment (melt/combined)	246% / 271%	222.7% / 247.7%
Total daily thermal energy	1081.5 MJ	990.7 MJ (−8.4%)
Average power demand	12.5 kW	11.47 kW (−8.2%)
Peak instantaneous demand	Not evaluated	36.61 kW
Branch valve count	8	6
Max concurrent priorities	2	3

11 Architecture II Overview: Adaptive Demand Allocation Architecture (ADA)

As of this revision, the plant's control logic is formalised under a single name — the **Adaptive Demand Allocation Architecture (ADA)** — resting on four pillars, restated here in their current, fully updated form.

11.1 Pillar 1: The Need-Array Priority Framework

Active demand is evaluated as a parallel need array across three tiers: *unbounded, guaranteed* (T_5, P1), *bounded, operator-supervised* (PO tank, P2), and *continuous, dedicated* (palm oil melting, P3 — now permanently allocated two branches rather than competing for shared time). As of this revision, **all three tiers can be genuinely concurrent**: the trunk, capped at its own safe $1.2\,\mathrm{m\,s^{-1}}$, serves whichever combination of branches has live demand, throttling flow rather than excluding a process (Section 7).

11.2 Pillar 2: Demand-Driven Dynamic Thermal Sizing

Unchanged in principle from Revision II: aggregate instantaneous demand $E_{\text{total}} = E_\alpha + E_5 + E_{C^*}$ is evaluated on a 10 s refresh window and converted to a dynamic water target, anchored between system idle (40 °C) and T_5's own worst-case draw. This revision adds a second sizing check operating on a different timescale: the **peak instantaneous coincidence** analysis of Section 9, which stress-tests the auxiliary array's rated capacity against the worst physically-plausible simultaneous demand across all three duties at once.

11.3 Pillar 3: Load-Responsive Operational Postures

Unchanged: T_5's own level (α_S) governs a four-tier hysteresis ladder (Eco/Efficiency/Standard/Full Throttle) for P3's operating posture, with mode selection feeding E_{total} causally rather than competing with it.

11.4 Pillar 4: Layered Operator-Trust Governance

Unchanged: the PO tank's suspicion meter escalates on repeated $2 \times n_d$ overrun events rather than cumulative daily volume, with autonomous isolation reserved for the third such event within a 24-hour window.

Architectural impact, updated. With the coil bore reduced and the need array expanded to true three-way concurrency, ADA now delivers ≈247.7 % of the original 2 t/day mandate — down from Revision II's 271 %, but achieved with 2 fewer control valves and a control architecture that no longer needs to discard any priority under contention. The isolated three-way flow-split penalty confirms that this concurrency gain was essentially free in thermal terms; the throughput given up this revision was spent almost entirely on the bore reduction, not on adding the third concurrent priority.

12 PLC Valve Mapping: Palm Oilen Flow Logic

12.1 Valve Identity

Table 7. Valve numbering and type.

Tag range	Count	Type / assignment
SV100 – SV105	6	Solenoid (water routing): SV100=T_5, SV101=TA, SV102=TB, SV103=TC, SV104/SV105=solar selector pair
PBV100 – PBV103	4	Pneumatic (oil dispense): one per tank, T_5/TA/TB/TC

SV and PBV numbering ranges overlap (e.g. "100" appears in both) but refer to physically distinct valves on separate lines with separate PLC output terminals — this is a shared tank-identifier convention, not a shared device.

12.2 Solar Selector Pair (SV104/SV105)

Exactly one of SV104/SV105 is closed at any time, never both, never neither: SV104 (Y15) closed → through-solar cycle; SV105 (Y16) closed → bypass-solar. This is the hardware implementation of the θ_{Sol} vs. T_{water} gating decision established in Architectural Revision II: $\theta_{\text{Sol}} > T_{\text{water}} \rightarrow$ through-solar; $\theta_{\text{Sol}} < T_{\text{water}} \rightarrow$ bypass.

12.3 Water Routing Table (Solenoid)

NOV (number of valves) = (explicit closed-valve count) +1, counting the SV104/SV105 pair as a single unit — confirmed against every row below with no exceptions.

Table 8. Water routing: line condition, valves closed, outputs energised.

#	Condition	NOV	Closed valves
1	Water T_5	4	101, 102, 103, (104,105)
2	Water TA	4	100, 102, 103, (104,105)
3	Water TB	4	100, 101, 103, (104,105)
4	Water TC	4	100, 101, 102, (104,105)
5	Water T_5+TA	3	102, 103, (104,105)
6	Water T_5+TB	3	101, 103, (104,105)
7	Water T_5+TC	3	101, 102, (104,105)
8	Water TA+TB	3	100, 103, (104,105)
9	Water TA+TC	3	100, 102, (104,105)
10	Water TB+TC	3	100, 101, (104,105)
11	Water TA+TB+TC	2	100, (104,105)
12	Water TA+TB+T_5	2	103, (104,105)
13	Water TA+T_5+TC	2	102, (104,105)
14	Water T_5+TB+TC	2	101, (104,105)

12.4 Oil Distribution Table (Pneumatic)

Table 9. Oil distribution: output port, valves closed, outputs energised.

#	Port	NOV	Closed valves
15	Oil x	2	PBV101, (104,105)
16	Oil y	3	PBV100, PBV103, (104,105)
17	Oil z	3	PBV100, PBV102, (104,105)

On the x/y/z port logic. Which valves must close to supply each output port (x, y, z) has been confirmed directly and cross-checked against the project's own P&ID; the mapping above is taken as accurate as given and is not re-derived from first principles in this document.

13 VFD Control

Both VFDs are controlled via discrete multi-speed (preset) digital inputs rather than RS485/Modbus, given that each requires only a small, fixed number of operating states:

- **VFD-01 (trunk):** 3 states (Off / LSM / HSM / FSM), each mapped to a pre-programmed target frequency in the VFD's own parameters, selected via dedicated, mutually-interlocked PLC digital outputs.
- **VFD-02 (oil pump):** 2 states (Off / On-at-safe-speed), a single pre-programmed frequency ceiling chosen to keep oil-side pipe velocity within safe bounds, selected via one digital output.

Per the VFD manufacturer's terminal documentation, confirm whether RUN/FWD is a command independent of speed-select, or combined into the same terminal logic, before finalising the interlock wiring — this determines whether 2 or 3 discrete lines are required for VFD-01.

14 PLC Implementation: Processes and Logic to Insert

This section consolidates every process, register, timer, alarm, I/O point, and control block required to implement the architecture described in this document on the PLC. Digital I/O is taken directly from the project's own AutoCAD Electrical panel schematic (sheet 6 of 10) and is presented as confirmed fact, not reconstructed. **Analog I/O is not yet included at the tag level** — the schematic notes that all analog inputs are detailed on a separate Analog Card sheet, not yet reviewed; the established channel *counts* (12 AI, 2 AO — Section "Hardware Bridging") still apply, but individual tag numbers are pending.

14.1 Analog Inputs (AI)

Eleven of the twelve confirmed analog input channels (Section "Hardware Bridging") are accounted for below. As with the digital I/O, real Coolmay tag numbers are not yet available — the analog card schematic has not been reviewed — so channels are labelled generically (AI-1 through AI-11) pending that confirmation.

Table 10. Analog input channel assignment.

Channel	Correspondence	Signal	Register
AI-1	Level — T_5	4-20mA (ultrasonic)	α_S
AI-2	Level — TA	4-20mA (ultrasonic)	α_{10A}
AI-3	Level — TB	4-20mA (ultrasonic)	α_{10B}
AI-4	Level — TC	4-20mA (ultrasonic)	α_{10C}
AI-5	Temperature — T_5	4-20mA (Pt100 transmitter)	θ_S
AI-6	Temperature — TA	4-20mA (Pt100 transmitter)	θ_{10A}
AI-7	Temperature — TB	4-20mA (Pt100 transmitter)	θ_{10B}
AI-8	Temperature — TC	4-20mA (Pt100 transmitter)	θ_{10C}
AI-9	Temperature — Solar loop	4-20mA (Pt100 transmitter)	θ_{Sol}
AI-10	Temperature — Thermal battery	4-20mA (Pt100 transmitter)	θ_H
AI-11	Flow — Trunk/outlet	4-20mA	Flow rate

Pending items. Real channel-to-terminal addressing awaits the Analog Card schematic; AI-1 through AI-11 are placeholder labels for this document only, not PLC tag names to be programmed as-is.

14.1.1 Reading AI Channels in the PLC

Toolchain assumption. The pattern below is written in Mitsubishi FX-series-style instructions (`FROM`/`TO`), programmed via GX Works2. This assumes the Coolmay L01S accepts an FX3U-compatible instruction set, consistent with how some budget PLC vendors position their hardware — but this has **not been confirmed** against Coolmay's own documentation. If the L01S uses its own proprietary software/instruction set instead, the two-step *read-then-scale* **logic** below still applies, but the exact instruction mnemonics will differ and should be substituted from Coolmay's own manual.

Each analog channel is handled in two steps: read the raw ADC count from the module's buffer memory into a holding register, then scale that raw count to a meaningful engineering-unit value in a second register.

Step 1 — Read raw value. Use `FROM` (read *from* an intelligent/special module into the CPU), not `TO` (which *writes to* the module — typically used only for configuration, not for reading sensor data).

```
FROM K0 K10 D0 K1
     |   |   |  |
     |   |   |  +-- number of words to transfer (1)
     |   |   +----- buffer memory start address (PLACEHOLDER)
     |   +--------- special/intelligent unit number (PLACEHOLDER)
     +------------- FROM = read from module into CPU
```

Step 2 — Scale to engineering units. General form:

$$\text{EngValue} = \text{RawCount} \times \frac{\text{Range}_{\text{max}}}{\text{ADC}_{\text{max}}} \tag{19}$$

```
MUL D0  K100  D2      // D2 = RawCount x Range_max  (PLACEHOLDER constant)
DIV D2  K4000 D4      // D4 = EngValue, scaled       (PLACEHOLDER constant)
```

Every constant above is a placeholder, not a value to program as-is:

- `K0` (unit number) and `K10` (buffer memory address) — depend entirely on which specific Coolmay AD module is installed and its position in the I/O chain. Must be taken from that module's own buffer-memory map, not assumed.
- `K4000` (ADC full-scale count) — assumes a 12-bit-class module (0–4000 count range, a common convention). Confirm the actual resolution of the installed module before use; some modules use 0–32000 or other ranges entirely.
- `K100` (engineering-unit range) — a worked example only (illustrating a 0–100 range, e.g. °C). Must be replaced per-channel: temperature channels, level channels, and the flow channel (if analog) each have their own real-world range and must be scaled independently.
- **A further subtlety not shown above:** whether raw count 0 corresponds to 4 mA or 0 mA depends on how each channel is configured on the module (some modules auto-zero a 4-20mA span to a 0-count baseline, others report the full 0-20mA span with a nonzero raw count at 4 mA). This must be confirmed per channel before trusting the scaled value, or every 4-20mA reading in this table will carry a silent offset error.
- `D0`, `D2`, `D4` — arbitrary register choices for this illustration only; must be coordinated against the full PLC register map once one exists, to avoid collision with registers already in use elsewhere in this document (Section "Calculated Registers").

14.2 Digital Inputs (DI)

Table 11. Digital input tags, panel schematic sheet 6/10.

Tag	Correspondence	Role
X0	E-STOP	Master safety
X1	Automation Enable/Disable	Master mode select
X2	Circulation Pump (manual)	VFD-01 / trunk manual button
X3	Oil Distribution Pump (manual)	VFD-02 manual button
X4	Jacket Pump (manual)	Jacketed-pipe circulation manual button
X5	AODD Pump (manual)	$T_{10} \rightarrow T_5$ transfer manual button
X6	Float — Heating tank (HEA), NC	Low-level fault input
X7	Float — Solar tank (SOL), NC	Low-level fault input
X10	Loop Exit	Manual-mode solar/bypass override

14.2.1 X10 — Loop Exit

Active **only while Automation is Disabled** (X1 = OFF). While Automation is Enabled, the automatic θ_{Sol} vs. T_{water} comparison (Section 14.12) governs SV104/SV105 exactly as before, and X10 has no effect.

```
IF Automation Disabled (X1 = OFF):
    IF X10 = ON  -> close SV104 (Y15), SV105 open  -> through-solar; Y25 = ON
    IF X10 = OFF -> close SV105 (Y16), SV104 open  -> bypass;       Y25 = OFF

IF Automation Enabled (X1 = ON):
    Automatic theta_Sol vs T_water comparison governs SV104/SV105
    Y25 = ON whenever SV104 is the closed valve (mirrors actual state)
```

14.2.2 X6/X7 — Float Switches (NC, Warning Only)

Both float switches are normally-closed, wired to the standard fail-safe convention: contact closed while level is adequate, contact opens on low level *or* on a wiring/sensor fault — so a broken connection reads identically to a genuine low-level condition rather than silently reading as fine.

```
X6 = ON  (contact closed) -> HEA level OK
X6 = OFF (contact open)   -> HMI warning: "HEA water loop level low"

X7 = ON  (contact closed) -> SOL level OK
X7 = OFF (contact open)   -> HMI warning: "SOL water loop level low"
```

Deliberately warning-only, no automated response, for now. On either float going low, the system raises a standing HMI warning and takes no further action — no pump interlock, no forced bypass. This is a deliberate scope limit, not an oversight: the current build authorization covers detection and operator notification only. **Deferred future work:** automated top-up logic, drawing from a reservoir tank to replenish the affected water loop, is explicitly planned for a later revision once clearance for that automation is granted — it should not be built ahead of that clearance.

Confirm NC polarity against the actual float switch datasheet before commissioning. "NC" is occasionally specified by manufacturers relative to the opposite resting position; the fail-safe, open-on-fault convention assumed above is standard for this application but should be verified, not assumed, against the specific unit installed.

14.3 Digital Outputs (DO)

Table 12. Digital output tags, panel schematic sheet 6/10.

Tag	Correspondence	Role
Y0	E-STOPPED	Master safety status lamp
Y1	Automation Enabled	Master mode status lamp
Y2	Circulation Pump (HSM)	Trunk: 2-branch speed select
Y3	Oil Distribution Pump (Active)	VFD-02 status lamp
Y4	Jacket Pump (Active)	Status lamp
Y5	AODD Pump (Active)	Status lamp
Y6	Circulation Pump (LSM)	Trunk: 1-branch speed select
Y7	Heating Element 1	Auxiliary immersion element (12 kW)
Y10	Heating Element 2	Auxiliary immersion element (12 kW)
Y11	SV100	T_5 branch valve
Y12	SV101	TA branch valve
Y13	SV102	TB branch valve
Y14	SV103	TC branch valve
Y15	SV104	Solar selector — through-solar
Y16	SV105	Solar selector — bypass-solar
Y17	Circulation Pump (FSM)	Trunk: 3-branch (full supply) speed select
Y20	Heating Element 3	Auxiliary immersion element (12 kW)
Y21	PBV100	T_5 oil valve
Y22	PBV101	TA oil valve
Y23	PBV102	TB oil valve
Y24	PBV103	TC oil valve

Y7/Y10/Y20 confirm the auxiliary array at 3×12 kW (36 kW total), matching the base VCH proposal's original figure and therefore the 36 kW ceiling used in the peak-wattage analysis (Section 9) without adjustment. This resolves an earlier Y20 naming conflict against a proposed "Solar Cycle Active" indicator — Y20 remains Heating Element 3, and the indicator, if still wanted, requires its own unused Y-tag rather than reusing Y20.

14.4 Master Safety and Mode Control: Ladder Logic

Safety-engineering caveat — read before implementing. The logic below assumes E-STOP is realised as a **hardwired safety relay or contactor circuit** that cuts power to pumps/motors directly, entirely independent of the PLC scan cycle — with X0 wired in purely as a **status/monitoring input**, so the PLC can display and log the E-Stopped state, not as the mechanism that actually performs the stop. Relying on PLC-scanned logic as the *sole* means of emergency stopping is not standard safe machinery practice (see IEC 60204-1): a hung scan, program fault, or communications delay would defeat a PLC-only implementation. **Confirm a hardwired safety circuit exists upstream of X0 before treating the logic below as sufficient.**

14.4.1 E-STOP (X0/Y0)

X0 is wired NC, consistent with every other safety-relevant input in this document (Section 14.2): contact closed in the normal state, open on either a genuine E-Stop or a wiring fault.

```
LDI X0
SET M1_ESTOP          // latches on X0 going OFF (pressed or fault)

LD  X_EStopReset       // PLACEHOLDER -- reset input not yet defined
RST M1_ESTOP           // clears the latch only on a deliberate reset

LD  M1_ESTOP
OUT Y0                 // Y0 lamp = E-Stopped

LDI M1_ESTOP
OUT M2_SystemOK        // "not E-Stopped" -- ANDed into every output
                        // rung in the program, manual and automatic alike
```

Open item: what clears the latch. No dedicated E-STOP reset input has been defined yet. Standard practice requires a *deliberate* re-arm action separate from simply releasing/twisting the palm button back out — otherwise the system could silently resume the instant the button pops back, without the operator confirming the area is actually clear. Confirm whether a separate physical Reset button exists (assign it a tag), or whether X1 (Automation Enable) toggling is meant to double as the reset action.

14.4.2 Automation Enable/Disable (X1/Y1)

```
LD  X1
OUT Y1                  // status lamp mirrors X1 directly

LD  X1
OUT M3_AutoEnabled       // flag consumed by every gated rung below
```

14.4.3 Gating Pattern: Manual vs. Automatic, Mutually Exclusive

Every output in the program — manual pump buttons and automatic arbiter/gating logic alike — is gated by `M2_SystemOK` first, then split on `M3_AutoEnabled` so manual and automatic paths can never both drive the same coil at once.

```
// Manual path -- live only when System OK AND Automation Disabled
LD  M2_SystemOK
ANI M3_AutoEnabled
AND X2                   // e.g. Circulation Pump manual button
OUT [pump output]
```

```
// Automatic path -- live only when System OK AND Automation Enabled
LD  M2_SystemOK
AND M3_AutoEnabled
AND [arbiter / need-array condition]
OUT [pump output]
```

This pattern repeats identically for every pump (Section on Pump Control Blocks) and every valve/gating block (need-array arbiter, solar selector, oil distribution) — `M2_SystemOK` and `M3_AutoEnabled` are the two conditions every real-world output coil in the program should ultimately trace back to.

Open item: behaviour on a mid-operation mode switch. If an operator toggles X1 while a pump is running automatically (e.g. mid- transfer, mid-dispense), does the affected process stop abruptly and wait for manual re-command, or does it need to complete its current step before control transfers? This affects valve/pump wear and, for P2 in particular, could interrupt a live dispense mid-stream — worth a decision before this is finalised, not left to fall out of whatever the mutual- exclusion logic happens to do by default.

14.5 Calculated Registers: Ladder Logic

Fixed-point arithmetic assumed throughout. Native floating-point instruction support for the L01S has not been confirmed (Section "Hardware Bridging"). Every calculation below uses scaled-integer arithmetic (e.g. percentages as plain integers, kW as integers, temperatures in tenths of a degree) rather than assuming a `real`/float data type is available. If floating point is later confirmed, these can be simplified.

$T_f(x, y, m_p)$ **is not yet resolved, despite the table's cross-reference.** The mass terms ($m_w = 90\,\text{kg}$, $m_{\text{ss}} = 205\,\text{kg}$) do not reconcile against the coil geometry alone at either the old 2-inch or new 1.5-inch bore — they almost certainly describe a larger assembly (T_5's jacket, a buffer volume, header piping) not yet identified. **This must be answered before Section 14.6 contains a real, ladder-ready equation**; it is not included below.

14.5.1 10-Second Refresh Oscillator

Drives E_α, E_5, E_{C^*}, E_{total}, and T_{water} recalculation. Standard self-resetting timer pattern — `T0` pulses for one scan every 10 s, then re-arms itself:

```
LDI T0
OUT T0 K100             // K100 = 100 x 0.1s = 10.0s
LD  T0
OUT M10_Refresh          // one-scan pulse, every 10s
```

14.5.2 T_{water} Target

Implements $T_{\text{water}} = 40 + 4.87P$ [kW], clamped 40–70 °C (Section **??**), in tenths of a degree for resolution. With P in whole watts:

```
LD  M10_Refresh
MUL D_Ptotal  K300   D200   // D200 = P[W] x 300
DIV D200      K6160  D202   // D202 = scaled increment, tenths of degree
ADD D202      K400   D204   // D204 = T_water x10, unclamped

LD< D204 K400
MOV K400 D204                // clamp lower bound (idle floor)

LD> D204 K700
MOV K700 D204                // clamp upper bound

MOV D204 D_Twater_target     // final, clamped, x10
```

14.5.3 Mode Register: Eco / Efficiency / Standard / Full Throttle

Single register `D_Mode` (0=Eco, 1=Efficiency, 2=Standard, 3=Full Throttle). Each rung only fires when `D_Mode` is currently in the state adjacent to the one being entered — this structurally enforces the hysteresis band from Section **??** and makes skipping a state or chattering between non-adjacent states impossible by construction, not just by convention.

```
// Eco(0) <-> Efficiency(1), band 90/85
LD= D_Mode K0
AND< D_AlphaS K85
MOV K1 D_Mode

LD= D_Mode K1
AND>= D_AlphaS K90
MOV K0 D_Mode

// Efficiency(1) <-> Standard(2), band 70/65
LD= D_Mode K1
AND< D_AlphaS K65
MOV K2 D_Mode

LD= D_Mode K2
AND>= D_AlphaS K70
MOV K1 D_Mode

// Standard(2) <-> Full Throttle(3), band 50/45
LD= D_Mode K2
AND< D_AlphaS K45
MOV K3 D_Mode
```

```
LD= D_Mode K3
AND>= D_AlphaS K50
MOV K2 D_Mode
```

Open item: power-up default. `D_Mode` needs a defined safe starting value before the first valid α_S reading arrives (e.g. default to Standard or Full Throttle as a conservative assumption on cold start) — not yet specified.

14.5.4 Suspicion Score ($S \in \{0, 1, 2, 3\}$)

```
// Increment on each confirmed 2xn_d overrun event (Section 3.2)
LD  M_Overrun_Pulse
INC D_Suspicion

LD> D_Suspicion K3
MOV K3 D_Suspicion          // cap at 3, no overflow

LD= D_Suspicion K1
OUT M_HMIPrompt
LD= D_Suspicion K2
OUT M_Alarm
LD= D_Suspicion K3
OUT M_DisableFeed
LD= D_Suspicion K3
OUT M_Alarm

// 5-min no-response auto-increment
LD  M_PromptActive
OUT T5  K3000                // 5 min = 300s
LD  T5
INC D_Suspicion
RST M_PromptActive

// 24hr reset -- PLACEHOLDER, tied to RTC/day-rollover, not detailed here
LD  M_DailyReset_Pulse
MOV K0 D_Suspicion
```

14.5.5 Auxiliary Stage Register (0/1/2/3)

Drives HE1/HE2/HE3 (Y7/Y10/Y20) staging, per the proposed thresholds from the staged-heating discussion (26/22, 14/10, 0/2 kW hysteresis bands — these three thresholds are proposed values, not yet confirmed).

```
// Stage transitions, same current-state pattern as the mode register
LD= D_AuxStage K0
AND> D_Ereq_kW K0
MOV K1 D_AuxStage
```

```
LD= D_AuxStage K1
AND<= D_Ereq_kW K2
MOV K0 D_AuxStage

LD= D_AuxStage K1
AND>= D_Ereq_kW K14
MOV K2 D_AuxStage

LD= D_AuxStage K2
AND< D_Ereq_kW K10
MOV K1 D_AuxStage

LD= D_AuxStage K2
AND>= D_Ereq_kW K26
MOV K3 D_AuxStage

LD= D_AuxStage K3
AND< D_Ereq_kW K22
MOV K2 D_AuxStage

// Drive elements from stage (cascading >= comparisons)
LD>= D_AuxStage K1
OUT Y7                    // HE1
LD>= D_AuxStage K2
OUT Y10                    // HE2
LD>= D_AuxStage K3
OUT Y20                    // HE3
```

14.5.6 Registers Already Covered Elsewhere

- **PO tank designation** — fully laddered in Section 14.10 (D51, M0–M3).
- **Need-array active count** — implemented as the LSM/HSM/FSM arbiter, Section 14.7.
- $S_{\mathbf{casual}}$, $S_{\mathbf{immediate}}$ — scoring formulas unchanged from the base VCH framework; the only new wiring is that the active PO-designation flags (M1/M2/M3, Section 14.10) now determine which tank is excluded from the ranking at runtime, rather than the exclusion being fixed to a hardcoded tank.

14.6 Thermal Blending Register: T_f Substitution

The coil's water and stainless-steel thermal mass are fixed by pipe geometry alone — 1.5" NPS, Schedule 10S, SS304, 10 m length — and do not vary at run time, so they collapse into a single constant coefficient rather than being recomputed on each scan cycle:

$$m_w = \rho_w \cdot \frac{\pi}{4} D_i^2 L \approx 14.34\,\text{kg}$$
$$m_{\text{ss}} = \rho_{\text{ss}} \cdot \frac{\pi}{4} \left(D_o^2 - D_i^2\right) L \approx 31.66\,\text{kg}$$
$$A \equiv m_w c_w + m_{\text{ss}} c_{\text{ss}} \approx 75\,808\,\text{J/K}$$

with $D_o = 0.048\,26\,\text{m}$, $D_i = 0.042\,723\,\text{m}$, $\rho_w = 1000\,\text{kg/m}^3$, $\rho_{\text{ss}} = 8000\,\text{kg/m}^3$, $c_w = 4184\,\text{J/kg}\cdot\text{K}$, $c_{\text{ss}} = 500\,\text{J/kg}\cdot\text{K}$.

Fully substituted T_f. With A fixed, the only run-time variable remaining is m_p (product/palm-oil mass in the blend), and the two temperature inputs are the heating supply water temperature θ_H and T_5's own current temperature θ_S:

$$T_f(\theta_H, \theta_S, m_p) = \frac{A\,\theta_H + m_p c_p\,\theta_S}{A + m_p c_p} = \frac{75808\,\theta_H + 2000\,m_p\,\theta_S}{75808 + 2000\,m_p}, \qquad c_p = 2000\,\text{J/kg}\cdot\text{K}$$

This is the form to implement directly in the PLC's T_f register — A is a hard-coded constant, not a computed register, and θ_H, θ_S, m_p are the only run-time inputs.

Variable assignment not independently confirmed. θ_H (as x) and θ_S (as y) are inferred from the role of T_f as a T_5 safety cutoff — blending incoming supply heat against the tank's current state — not from an explicit definition given elsewhere in this document. Confirm this assignment (or its reverse) before this equation is finalised in the register table, since it directly drives the P1 interlock threshold in Section 14.6 above.

Schedule assumption. A above assumes Schedule 10S wall thickness (0.109") for the 1.5" NPS coil, per the confirmed bore-reduction upgrade. If the as-built coil uses a different schedule (e.g. Sch 40S), D_i and therefore A change — cross-check against the schedule called out in the coil specification section before this constant is finalised in the register table.

This register cannot use the 16-bit pattern established in Section 14.5. $A = 75\,808\,\text{J/K}$ alone exceeds the 16-bit signed integer range (-32768 to 32767) used for every other calculated register in this document. Every step below therefore uses **32-bit (double-word) instructions** (`DMUL`, `DADD`, `DDIV`) rather than their 16-bit counterparts (`MUL`, `ADD`, `DIV`) — this is a deliberate, necessary departure from the rest of Section 14.5, not an inconsistency.

14.6.1 Registers

Table 13. T_f ladder registers.

Tag	Purpose
D_ThetaH	θ_H, tenths °C (AI-10 scaled, Section 14.1)
D_ThetaS	θ_S, tenths °C (AI-5 scaled, Section 14.1)
D_Mp	m_p, whole kg — **source not yet defined, see below**
D330:D331 – D373	Working 32-bit register pairs, this block only
D_Tf	Final result, tenths °C, 16-bit (safe: result is a temperature, well within 16-bit r
M_TfInterlock	ON when $T_f \geq 55\,°C$

Working register range (D330–D373) is illustrative only, chosen to avoid the AI scaling block (D0–D4) and the PO designation block (D50–D51) — must be coordinated against the complete register map once one exists, same caveat as every other placeholder address in this document.

14.6.2 Step 0 – Zero-Extend the 16-bit Sources

θ_H, θ_S, and m_p live in ordinary 16-bit registers (Section 14.1). Passing a single 16-bit register directly into a D-prefixed instruction silently pulls in the *next* register as the unintended high word — a common and easy-to-miss source of corrupted results. To avoid this, each value is copied into a dedicated low word with its high word explicitly zeroed, every cycle, before use:

```
MOV D_ThetaH D330   // D330:D331 = theta_H, zero-extended
MOV K0       D331

MOV D_ThetaS D332   // D332:D333 = theta_S, zero-extended
MOV K0       D333

MOV D_Mp     D334   // D334:D335 = m_p, zero-extended
MOV K0       D335
```

14.6.3 Step 1 – Two Products

```
DMUL K75808 D330 D340   // D340:D341(:D342:D343) = A x theta_H
DMUL D334   K2000 D344   // D344:D345(:D346:D347) = m_p x c_p
```

14.6.4 Step 2 – Second Product, Chained

```
DMUL D344 D332 D350      // D350:D351(:D352:D353) = (m_p*c_p) x theta_S
```

14.6.5 Step 3 – Numerator and Denominator

```
DADD D340 D350 D360      // D360:D361 = Numerator = A*theta_H + m_p*c_p*theta_S
DADD K75808 D344 D362     // D362:D363 = Denominator = A + m_p*c_p
```

14.6.6 Step 4 – Divide, Then Interlock

```
DDIV D360 D362 D370      // D370:D371 = quotient = Tf, tenths degC
MOV  D370 D_Tf             // 16-bit result register

LD>= D_Tf K550             // 55.0 degC x10
OUT  M_TfInterlock

LD   M_TfInterlock
OUT  M4_ThrottleOrDivert   // feeds P1's "throttle or divert flow" action --
                            // coordinate with the actual valve/pump rungs
                            // that implement that response
```

Magnitude check, so the 32-bit path doesn't silently overflow either. With $\theta_H, \theta_S \leq 700$ (tenths, i.e. 70 °C) and m_p up to roughly 1000 kg: the largest intermediate product ($m_p c_p \theta_S \approx 1000 \times 2000 \times 700 \approx 1.4 \times 10^9$) and the numerator sum ($\approx 1.45 \times 10^9$) both stay under the 32-bit signed ceiling ($\approx 2.147 \times 10^9$) with headroom. **This headroom shrinks if m_p can exceed roughly** 1500 kg — worth a sanity check once the real maximum batch/product mass is known.

Open item: m_p has no defined source register or acquisition method. Every other input to this block traces to a confirmed AI channel (Section 14.1) or an existing calculated register. m_p does not — it is not produced anywhere else in this document. Before `D_Mp` can be wired to anything real, it needs to be pinned down: is it operator-entered on the HMI (like n_d in P2), derived from a level-change calculation on the source tank, or fixed at a conservative worst-case constant for interlock purposes? This is not a formatting gap — without it, D_Mp is an empty register and the whole block is inert.

Open item: refresh rate mismatch with the rest of this section. Every other calculated register in Section 14.5 is gated behind `M10_Refresh`, the 10 s pulse. P1 (Section 14) instead specifies T_f should be *"continuously"* evaluated. As drafted above, the block runs every scan (unconditional), consistent with P1's wording — but this is a genuine deviation from the 10 s pattern elsewhere, not an oversight, and should be confirmed as intentional given T_f is a safety interlock threshold rather than a slower-moving supervisory value like T_{water}.

Rounding direction. `DDIV` truncates toward zero, so the computed T_f can read up to 0.09 °C *below* the true value (never above). For a safety interlock that trips on $T_f \geq 55$ °C, this is the non-conservative direction — truncation could delay a trip by a fraction of a degree rather than trigger one early. The margin is small enough to likely be immaterial, but it should be an informed decision, not an unnoticed side effect of integer division.

Still carrying forward from Section 14.6: the θ_H/θ_S assignment in the equation itself is not independently confirmed. The ladder above implements $x = \theta_H$, $y = \theta_S$ as inferred there. If that assignment is reversed once confirmed, only `D330`/`D332` (and the corresponding `DMUL` operands in Step 1) need to swap — the rest of the block is unaffected.

14.7 Need-Array Arbiter: Three-Way Flow-Sharing

The trunk now operates in three confirmed hardware states — LSM (Y6), HSM (Y2), FSM (Y17) — one branch, two branches, and three branches active respectively. This supersedes the earlier two-branch-only, "discard the lowest priority" contention rule entirely: two of the three branches are permanently dedicated to T_{10} melting (P3), and the third is shared between P1 (T_5) and P2 (PO tank). Any time P1 or P2 has live demand, the trunk is necessarily in FSM — serving all three branches at once, each throttled below its individual 1.2 m s^{-1} design rate (Section 3) — rather than any priority being excluded.

Naming collision to watch for. FSM ("Full Supply Mode," a trunk flow-rate state, Y17) and Full Throttle (a P3 operating-posture mode from the hysteresis ladder, Section **??**) are **unrelated systems that happen to share the word "full."** A tank can be in Full Throttle mode while the trunk itself is in LSM, HSM, or FSM depending on what else is concurrently active — these must not be conflated in the PLC program or in future documentation.

Open item surfaced by implementing this block: does the trunk ever actually reach LSM? Section 14.7 states P3 (T_{10} melting) is **permanently** allocated two branches, with the third shared between P1/P2. If P3's two branches are always active whenever the plant is running at all, the baseline state is HSM (2 branches), and LSM (1 branch) has no describable trigger condition under the architecture as written — it would only occur if P3 itself can idle down to a single branch, which is not stated anywhere in this document. The ladder below implements LSM as "P3 running, but only one of its two branches has live demand," as the closest available reading, but this should be confirmed rather than assumed: if P3's two branches are truly inseparable, LSM may be an unreachable state in practice and Y6 may never energise.

14.7.1 Registers and Flags

Table 14. Arbiter registers and flags.

Tag	Purpose
M_P1Demand	P1 (T_5) live demand, from $\theta_S < 40\,^{\circ}\mathrm{C}$ (Section 14)
M_P2Demand	P2 (PO tank) live demand, dispense path currently open (Section **??**)
M_P3BranchA	P3 dedicated branch A has live demand
M_P3BranchB	P3 dedicated branch B has live demand
D_NeedCount	Need-array active count, 0–3

14.7.2 Step 1 – Count Active Branches

The shared third branch (P1/P2) counts as at most one, even if both P1 *and* P2 have live demand simultaneously — they contend for the same single shared branch, not two separate ones, consistent with Pillar 1 (Section 11):

```
LD  M_P1Demand
OR  M_P2Demand
OUT M_SharedBranchLive     // shared branch: P1 OR P2, never counted twice

MOV K0 D_NeedCount

LD  M_P3BranchA
INC D_NeedCount
LD  M_P3BranchB
INC D_NeedCount
LD  M_SharedBranchLive
INC D_NeedCount
```

14.7.3 Step 2 – Map Count to Trunk State

Mutually-interlocked outputs, per the VFD documentation caveat already on record (Section 13) – exactly one of Y6/Y2/Y17 energised whenever the trunk is running at all:

```
LD  M2_SystemOK
AND M3_AutoEnabled
AND= D_NeedCount K1
OUT Y6                     // LSM -- see open item above

LD  M2_SystemOK
AND M3_AutoEnabled
AND= D_NeedCount K2
OUT Y2                     // HSM -- baseline, P3's two branches only

LD  M2_SystemOK
```

```
AND M3_AutoEnabled
AND= D_NeedCount K3
OUT Y17                    // FSM -- P3 (both branches) + shared branch live
```

This directly implements the resultbox claim in Section 14.7: any time `M_SharedBranchLive` is ON (P1 or P2 has live demand), `D_NeedCount` can only be 2 or 3 depending on how many P3 branches are simultaneously live – and per the architecture, both P3 branches are supposed to be permanently live, which forces `D_NeedCount=3` and therefore FSM, matching "Any time P1 or P2 has live demand, the trunk is necessarily in FSM" exactly.

14.7.4 Step 3 – Manual Path (Automation Disabled)

Follows the same gating pattern as every other pump block (Section 14.4): with automation disabled, X2 drives the trunk directly, and the arbiter above does not run.

```
LD  M2_SystemOK
ANI M3_AutoEnabled
AND X2
OUT Y2                     // manual: single fixed speed (HSM), no
                            // LSM/FSM distinction available manually
```

Open item: manual-mode speed selection. The manual path above defaults X2 to HSM only, since the physical panel (Section 14) has a single trunk pump button, not a 3-position selector. If an operator needs LSM or FSM specifically while in manual mode (e.g. during commissioning or a fault on the automatic arbiter), there is currently no way to select it from the panel as specified – confirm whether this is acceptable or whether manual mode needs its own speed-select input.

14.8 Mode State Machine

Implements the hysteresis ladder of Section **??**: four states (Eco, Efficiency, Standard, Full Throttle), transitions gated by the rising/falling thresholds in that section's table, each with a 5 % deadband. The mode register feeds P3's operating rate setpoint, which in turn feeds E_α (Section **??**). Independent of, and not to be confused with, the trunk's LSM/HSM/FSM state (Section 14.7).

Ladder logic already implemented. The full `D_Mode` ladder — registers, the six hysteresis-band rungs (90/85, 70/65, 50/45), and the current-state-gated structure that makes chattering or state-skipping structurally impossible — is given in Section 14.5, subsubsection "Mode Register: Eco / Efficiency / Standard / Full Throttle." Not repeated here to avoid two copies of the same logic drifting apart if one is edited later.

Still open (carried from Section 14.5): `D_Mode` has no defined power-up default before the first valid α_S reading arrives.

14.9 P1 — T_5 Block

- $\theta_S < 40\,^{\circ}\text{C}$ triggers the P1 heat-demand flag (open SV100, request a branch from the arbiter).
- T_f interlock: continuously evaluate $T_f(x, y, m_p)$ (Section 14.6); if it exceeds 55 °C, throttle or divert flow.
- **Batch-release condition:** $\alpha_S < 50\,\%$ **and** the highest-ranked non-PO T_{10} tank $\geq 36\,^{\circ}\text{C} \rightarrow$ energise the **AODD Pump (Y5)**, transferring molten oil from the selected tank to T_5.

14.9.1 Registers and Flags

Table 15. P1 ladder registers and flags.

Tag	Purpose
D_ThetaS	θ_S, tenths °C (AI-5, Section 14.1)
D_AlphaS	α_S, whole % (AI-1, Section 14.1)
D_RankedTankTemp	Temperature of the highest-ranked non-PO T_{10} tank, tenths °C – **source not**
M_P1Demand	P1 heat-demand flag (consumed by the arbiter, Section **??**)
M_BatchRelease	Batch-release condition met

14.9.2 Step 1 – Heat-Demand Flag

```
LD<  D_ThetaS K400        // 40.0 degC x10
OUT  M_P1Demand

LD   M_P1Demand
OUT  Y11                  // SV100, T5 branch valve
                            // M_P1Demand also feeds M_SharedBranchLive
                            // in the arbiter (Section~\ref{sec:arbiter-ladder})
```

14.9.3 Step 2 – T_f Interlock Action

Already laddered in full (Section **??**); `M_TfInterlock` is the output consumed here:

```
LD   M_TfInterlock
OUT  M4_ThrottleOrDivert   // as defined in Section~\ref{sec:tf-ladder} --
                            // coordinate with the specific valve/pump
                            // rungs that carry out the throttle/divert
                            // action once that response is designed
```

14.9.4 Step 3 – Batch-Release Condition

```
LD    D_AlphaS
SUB   K1
LD<   D_AlphaS K50          // alpha_S < 50%
AND>= D_RankedTankTemp K360  // ranked tank >= 36.0 degC
OUT   M_BatchRelease

LD    M_BatchRelease
OUT   Y5                     // AODD Pump
```

Open item: `D_RankedTankTemp` has no ladder source. Step 3 assumes this register already holds the temperature of "the highest-ranked non-PO T_{10} tank," but the S_{casual}/ $S_{\text{immediate}}$ ranking that determines *which* tank that is has never been laddered in this document – Section 14.5 only notes the scoring formulas are unchanged from the base VCH framework and that PO-designation flags (Section 14.10) now exclude the designated tank from ranking. Until that ranking block exists (compute $S_{\text{casual}}/S_{\text{immediate}}$ for TA/TB/TC, exclude the PO-designated one via M1/M2/M3, select the highest score, and read that tank's AI-6/7/8 temperature into `D_RankedTankTemp`), Step 3 above is not runnable as written – it is correct *given* that input, not a complete implementation.

Line 1–2 of Step 3 (`LD`/`SUB`) is dead code, left in deliberately to flag it. It computes a value that is immediately discarded and never referenced again – remove before this is programmed; kept visible here only so it doesn't silently vanish and get mistaken for something load-bearing elsewhere.

14.10 PO Tank Designation: HMI Selector and Ladder Logic

PO tank designation uses a single numeric register as the sole source of truth for which tank (if any) currently holds the designation — a standard pattern for exclusive equipment-assignment logic, chosen specifically because a single register makes an invalid state (e.g. two tanks simultaneously designated) structurally impossible, rather than something that has to be prevented by additional interlock logic.

Designation register convention (confirmed):

Value	Meaning
0	No PO tank designated — all three T_{10} tanks run palm oil
1	TA is the PO tank
2	TB is the PO tank
3	TC is the PO tank

Designation is optional — 0 ("None") is a valid, standing operating state, not merely a placeholder value.

14.10.1 Registers

Table 16. PO designation registers and flags.

Tag	Purpose
D50	Pending selection, written live by the HMI radio-group (0–3)
D51	Active designation — the value all downstream logic actually reads
M0	Active designation = 0 (no PO tank)
M1	Active designation = 1 (TA is PO)
M2	Active designation = 2 (TB is PO)
M3	Active designation = 3 (TC is PO)
X_Confirm	HMI confirm button (momentary)

Register numbers (D50/D51) are illustrative placeholders, chosen to avoid an obvious collision with the AI scaling registers (D0/D2/D4, Section 14.1) — final addresses must be coordinated against the complete register map once one exists.

14.10.2 Ladder Logic

Step 1 — HMI writes the pending selection. The HMI radio-group writes directly and continuously to D50 as the operator taps TA/TB/TC/None — this is a live *pending* value, not yet acted on.

Step 2 — Confirm commits the pending value, on a rising edge only. A plain continuous read of D50 would let a transient or accidental tap immediately reassign the designation; gating the commit to a single rising-edge pulse of the confirm button prevents that.

```
LD  X_Confirm
PLS M100                  // M100 pulses ON for exactly one scan,
                           // on the rising edge of X_Confirm only

LD  M100
MOV D50  D51               // commit: pending -> active, one-shot
```

Step 3 — Decode D51 into exclusive flags. Straight equality comparisons; exactly one flag is ever true, by construction.

```
LD= D51 K0   OUT M0        // no PO tank
LD= D51 K1   OUT M1        // TA is PO
LD= D51 K2   OUT M2        // TB is PO
LD= D51 K3   OUT M3        // TC is PO
```

14.10.3 Downstream Consumption

- **P2 block** (Section **??**) resolves its active SV/PBV pair from whichever of M1/M2/M3 is set — e.g. M1 ON → P2 operates SV101/PBV101 (TA). With M0 ON, P2 has no valid tank to operate and remains idle.
- **P3 scoring block** excludes whichever tank corresponds to the active flag from $S_{\text{casual}}/S_{\text{immediate}}$ ranking; with M0 ON, all three T_{10} tanks participate in scoring normally.
- **On any change of D51** (a designation change taking effect), the newly-released tank re-enters P3 scoring immediately and the newly-designated tank is immediately excluded — there is no transition delay or grace period in this logic.

No automated contamination interlock, by explicit decision. Changing designation does not check tank level, require an emptied-tank confirmation, or block on any physical condition — the operator's own manual valve is the sole safeguard against cross-contaminating product between designation changes. This is a deliberate, budget-driven scope decision, not an oversight, and should be documented as such rather than revisited as a "missing" interlock later.

Open item: access level on the confirm step. Whether X_Confirm requires only a standard operator tap or a supervisor PIN (mirroring the escalation pattern already used for the suspicion meter, Section **??**) has not yet been decided. Given there is no automated contamination check, a logged, attributable confirm action may be worth considering even without a full PIN gate — e.g. timestamping every D51 change to the trend log (Section 14.2) for traceability, since the operator is the sole safeguard here. This is offered as a lightweight suggestion, not a requirement, and remains open for a decision.

14.11 Pump Control Blocks

Table 17. Pump I/O summary.

Pump	Manual (X)	Status/Select (Y)	Auto-trigger condition
Circulation (trunk, VFD-01)	X2	Y6/Y2/Y17 (LSM/HSM/FSM)	Need-array active count (Section 14.7)
Oil Distribution (VFD-02)	X3	Y3 (Active)	Any of $\delta_X, \delta_Y, \delta_Z$ live
Jacket	X4	Y4 (Active)	Not modelled in this revision — included as a confirmed I/O point only
AODD (HPP)	X5	Y5 (Active)	P1 batch-release condition (Section above)

14.12 Solar and Auxiliary Gating Block

Compare θ_{Sol} against the dynamic T_{water} target: $\theta_{\text{Sol}} > T_{\text{water}} \rightarrow$ close SV104 (through-solar, Y15); $\theta_{\text{Sol}} < T_{\text{water}} \rightarrow$ close SV105 (bypass-solar, Y16) — exactly one of the two, always, per the fail-safe rationale of Section 12 (normally-open, energise-to-close, so a fault or power loss always leaves a flow path open and the pump can never dead-head). This automatic comparison is subject to revision pending the Loop Exit override/replace clarification flagged in the Digital Inputs table above.

Staged auxiliary heating. Rather than binary on/off control, HE1/HE2/HE3 (Y7/Y10/Y20) are staged discretely against $E_{\text{aux,required}}$, evaluated only while bypass-solar is the active driver:

$$E_{\text{aux,required}} = E_{\text{total}} \quad \text{(auxiliary is the active driver, i.e. bypass-solar engaged)}$$

Since each element is a fixed 12 kW, staging falls out in whole-element increments, with ±2 kW hysteresis bands around each threshold to prevent chatter (same principle as the T_5 mode ladder, Section **??**):

Table 18. Auxiliary staging thresholds (3-element array, contingent on the Y20/HE3 resolution above).

$E_{\text{aux,required}}$	Stage	Elements active
Rising ≥26 kW / falling <22 kW	3	HE1+HE2+HE3
Rising ≥14 kW / falling <10 kW	2	HE1+HE2
Rising >0 kW / falling ≤2 kW	1	HE1
≈0 kW	0	Off

Hysteresis margin not independently specified. The ±2 kW bands above were proposed to match the existing mode-ladder pattern, not supplied as a fixed design figure — flag if different margins are wanted before this is finalised in ladder logic.

The thermal battery recharge interlock operates as specified in the base VCH control architecture, unaffected by the staging change above.

14.13 Fault Detection Block

Implements the physics-based alarm philosophy established in Architectural Revision I: continuously compare each tank's *observed* heating/cooling rate against the *maximum physically achievable* rate given the coil's known delivery capacity and that tank's own computed standing-loss rate.

On detecting an unnatural rate: sound the alarm immediately; present a stop button on the HMI; if the operator presses stop, halt the affected process; if not, the system continues operating with the alarm latched as a standing notice — deliberately advisory, consistent with the operator-trust principle applied to P2 (Section **??**).

14.14 Timers

- Suspicion meter: 24 hr reset; 10 min no-HMI-response auto-increment.
- $E_{\text{total}}/T_{\text{water}}$ refresh: 10 s.
- Thermal recharge sequence: circulate-to-36 °C, then full-auxiliary.

14.15 Alarms and Interlocks

- E-STOP (X0) → global output kill, Y0 lamp latched.
- T_f boundary exceeded → T_5 interlock.
- $\theta_H < 60\,^{\circ}\mathrm{C}$ → recharge alarm sequence.
- T_{10} high-limit $> 75\,^{\circ}\mathrm{C}$ → valve close + alarm.
- Physics-based heating/cooling rate violation → alarm + operator stop/continue choice (Section 14.13).
- Suspicion meter S=2 (alarm), S=3 (disable + alarm).
- PO tank valve/designation mismatch: cross-check that an isolated tank's manual valve has not been reopened outside the designated dispense workflow.

14.16 Two Operator Interfaces: Physical Panel vs. Touchscreen HMI

This architecture has **two distinct operator interfaces**, not one. The **physical panel** (E-STOP, Automation Enable/Disable, and the four pump manual buttons with paired status lamps, X0–X5/Y0–Y6) handles master safety and manual pump override, confirmed directly from the schematic. The **touchscreen HMI** (Coolmay) handles everything process-level: PO tank designation, dispense quantity entry, suspicion meter prompts, mode indicator, the fault-alarm stop/continue button, and the I/O diagnostic screen below. The two do not overlap in function and should be documented/wired as separate systems.

- **Physical panel:** E-STOP, Automation Enable/Disable, 4 pump manual buttons + status lamps (X0–X5, Y0–Y6).
- **Touchscreen HMI:** PO tank designation control, dispense quantity entry (n_d), process-complete button, suspicion meter status/prompts, mode indicator (read-only, four-state), fault alarm stop/continue button, I/O diagnostic screen (below), trend logging for θ, α, E_{total}, T_{water}.

14.17 I/O Diagnostic Screen

A dedicated HMI troubleshooting screen displays live raw values across the full I/O map, laid out as a simple continuously-refreshed table: each AI channel's raw signal (mA and scaled engineering units), each DI's energised/de-energised state, and each DO's active/inactive state. This is standard commissioning/maintenance practice, touches no control logic, and sits alongside the other touchscreen HMI elements listed above.

15 System Estimation and Conclusions

1. **The bore reduction and the concurrency expansion are thermally separable, and only one of them is expensive.** The 2 inch→1.5 inch bore change costs 16–19 % of delivered heat across every duty; adding genuine three-way concurrency on top of that costs under 2 % further. The valve-count and control-complexity benefits of true three-way concurrency were, in thermal terms, essentially free.

2. **The system still clears the original mandate by** ≈ 2.5× (247.7 % combined), comfortably absorbing the ≈8.6 % throughput reduction this revision accepts.

3. **The peak instantaneous demand check is new to this revision and worth carrying forward.** At 36.61 kW against a 36 kW rated auxiliary array, the margin is narrow enough that it should inform the electrical design phase explicitly, even though the underlying scenario is a conservative worst-case coincidence rather than typical operation.

4. **The three-way need-array arbiter is architecturally simpler than Revision II's, not more complex**, despite serving more priorities at once — flow-sharing under a velocity cap replaces the discard-lowest-priority contention rule entirely, removing a decision branch from the PLC logic rather than adding one.

5. **PO tank governance and mode-selection logic (Pillars 3 and 4 of ADA) are unaffected by this revision** and carry forward unchanged from Architectural Revision II.

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